

Lecture-08 [20 August 2026]

Recap: A random variable X is said to be **sub-Gaussian** if any one of the following equivalent conditions are satisfied.

① There exists $K_1 > 0$ such that

$$P(|X| \geq \varepsilon) \leq 2 \exp\left(-\frac{\varepsilon^2}{K_1^2}\right) \quad \forall \varepsilon > 0.$$

② There exists $K_2 > 0$ such that

$$\left(E[|X|^p]\right)^{1/p} \leq K_2 \sqrt{p} \quad \forall p \geq 1.$$

③ There exists $K_3 > 0$ such that

$$E[e^{t^2 X^2}] \leq e^{t^2 K_3^2} \quad \forall t \in \left[-\frac{1}{K_3}, \frac{1}{K_3}\right].$$

④ There exists $K_4 > 0$ such that

$$E[e^{X^2/K_4^2}] \leq 2.$$

Additionally, if $E[X] = 0$:

⑤ There exists $K_5 > 0$ such that

$$E[e^{tX}] \leq e^{t^2 K_5^2} \quad \forall t \in \mathbb{R}.$$

Definition (sub-Gaussian norm)

The sub-Gaussian norm of a sub-Gaussian random variable X , denoted $\|X\|_{\psi_2}$, is defined as

$$\|X\|_{\psi_2} := \inf \{K > 0 : E[e^{X^2/K^2}] \leq 2\}.$$

Example: If $X \sim N(0, \sigma^2)$, what is $\|X\|_{\psi_2}$?

Exercise: Show that if $X \sim N(0, \sigma^2)$, then

$$E[e^{tX^2}] = M_{X^2}(t) = \begin{cases} \frac{1}{(1 - 2\sigma^2 t)^{1/2}}, & t < \frac{1}{2\sigma^2}, \\ +\infty, & \text{otherwise.} \end{cases}$$

Then, we have

$$\|X\|_{\psi_2} = \inf \left\{ K > 0 : \frac{1}{\left(1 - \frac{2\sigma^2}{K^2}\right)^{1/2}} \leq 2 \right\} \quad K^2 > 2\sigma^2$$

$$= \inf \left\{ K > 0 : 1 - \frac{2\sigma^2}{K^2} \geq \frac{1}{4} \right\}$$

$$= \inf \left\{ K > 0 : K^2 \geq \frac{8\sigma^2}{3} \right\} = \sqrt{\frac{8\sigma^2}{3}}.$$

Thus, $\|X\|_{\Psi_2} = \sqrt{\frac{8\sigma^2}{3}}$ if $X \sim N(0, \sigma^2)$.

SUB-EXPONENTIAL RANDOM VARIABLES

A random variable X is said to be **sub-exponential** if any of the following equivalent conditions hold.

- ① There exists a constant $K_1 > 0$ such that

$$\mathbb{P}(|X| \geq \varepsilon) \leq 2 \exp\left(-\frac{\varepsilon}{K_1}\right) \quad \forall \varepsilon > 0.$$

- ② There exists a constant $K_2 > 0$ such that

$$\left(\mathbb{E}[|X|^p]\right)^{1/p} \leq K_2 p \quad \forall p \geq 1.$$

- ③ There exists a constant $K_3 > 0$ such that

$$\mathbb{E}[e^{t|X|}] \leq e^{K_3 t} \quad \forall t \in \left[0, \frac{1}{K_3}\right].$$

- ④ There exists a constant $K_4 > 0$ such that

$$\mathbb{E}[e^{|X|/K_4}] \leq 2.$$

Additionally, if $\mathbb{E}[X] = 0$:

- ⑤ There exists a constant $K_5 > 0$ such that

$$\mathbb{E}[e^{tX}] \leq e^{K_5^2 t^2} \quad \forall t \in \left[-\frac{1}{K_5}, \frac{1}{K_5}\right].$$

Remark: If $\mathbb{E}[X] \neq 0$, one should replace X with $X - \mathbb{E}[X]$ in all the 5 conditions.

Definition (sub-exponential norm)

If X is a sub-exponential random variable, its sub-exponential norm, denoted $\|X\|_{\Psi_1}$, is defined as

$$\|X\|_{\psi_1} = \inf \{k > 0 : E[e^{|X|/k}] \leq 2\}.$$

Properties of sub-exponential norm:

① Exercise: $\|\alpha X\|_{\psi_1} = |\alpha| \|X\|_{\psi_1} \quad \forall \alpha \in \mathbb{R}.$

② Exercise: If X and Y are sub-exponential, then

$$\|X+Y\|_{\psi_1} \leq \|X\|_{\psi_1} + \|Y\|_{\psi_1}.$$

③ If X and Y are independent and sub-exponential, then

$$\|X+Y\|_{\psi_1} \leq \sqrt{\|X\|_{\psi_1}^2 + \|Y\|_{\psi_1}^2}.$$

Proof of ③: Without loss of generality, let $E[X] = 0 = E[Y]$.

Then, we know that

$$E[e^{tX}] \leq e^{c_1^2 t^2 \|X\|_{\psi_1}^2} \quad \begin{array}{l} \text{for some universal constant } c_1 > 0, \\ \text{for all } t \in \left[-\frac{1}{c_1 \|X\|_{\psi_1}}, \frac{1}{c_1 \|X\|_{\psi_1}} \right] \end{array}$$

$$E[e^{tY}] \leq e^{c_2^2 t^2 \|Y\|_{\psi_1}^2} \quad \begin{array}{l} \text{for some universal constant } c_2 > 0, \\ \text{for all } t \in \left[-\frac{1}{c_2 \|Y\|_{\psi_1}}, \frac{1}{c_2 \|Y\|_{\psi_1}} \right]. \end{array}$$

Taking $c = \max\{c_1, c_2\}$, and noting that

$$c \sqrt{\|X\|_{\psi_1}^2 + \|Y\|_{\psi_1}^2} \geq \max\{c_1 \|X\|_{\psi_1}, c_2 \|Y\|_{\psi_1}\},$$

we get

$$\begin{aligned} E[e^{t(X+Y)}] &\leq e^{c_1^2 t^2 \|X\|_{\psi_1}^2 + c_2^2 t^2 \|Y\|_{\psi_1}^2} \\ &\leq e^{c^2 t^2 (\|X\|_{\psi_1}^2 + \|Y\|_{\psi_1}^2)} \quad \forall t \in \left[-\frac{1}{c \sqrt{\|X\|_{\psi_1}^2 + \|Y\|_{\psi_1}^2}}, \frac{1}{c \sqrt{\|X\|_{\psi_1}^2 + \|Y\|_{\psi_1}^2}} \right] \end{aligned}$$

thus proving that

$$\|X+Y\|_{\psi_1} \leq \sqrt{\|X\|_{\psi_1}^2 + \|Y\|_{\psi_1}^2}.$$

④ If X is sub-Gaussian, then X is sub-exponential.
However, the converse is not true (eg: $X \sim \text{Exp}(\lambda)$)

Proof of (4) :

$$X \text{ sub-Gaussian} \Rightarrow \left(\mathbb{E}[|x|^p] \right)^{1/p} \leq \overset{\text{there exists } k_2 > 0}{k_2 \sqrt{p}} \quad \forall p \geq 1$$

$$\leq k_2 p \quad \forall p \geq 1$$

(5) If X is sub-Gaussian, then X^2 is sub-exponential, and

$$\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2.$$

Proof of (5) :

$$X \text{ sub-Gaussian} \Rightarrow \mathbb{E}\left[e^{X^2/\|X\|_{\psi_2}^2}\right] \leq 2.$$

Taking $Y = X^2$, it follows that

$$\mathbb{E}\left[e^{Y/\|X\|_{\psi_2}^2}\right] \leq 2 \Rightarrow Y \text{ is sub-exponential}$$

$$\|Y\|_{\psi_1} \leq \|X\|_{\psi_2}^2 \rightarrow \textcircled{a}$$

Also, we have

$$\mathbb{E}\left[e^{Y/\|Y\|_{\psi_1}}\right] \leq 2 \Rightarrow \mathbb{E}\left[e^{X^2/\|Y\|_{\psi_1}}\right] \leq 2$$

$$\Rightarrow \|X\|_{\psi_2} \leq \sqrt{\|Y\|_{\psi_1}} \rightarrow \textcircled{b}$$

Combining $\textcircled{a}$ and $\textcircled{b}$, we get

$$\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2.$$

(6) If X and Y are sub-Gaussian (and not necessarily independent), then XY is sub-exponential. Furthermore,

$$\|XY\|_{\psi_1} \leq \|X\|_{\psi_2} \|Y\|_{\psi_2}.$$

Proof of (6) :

$$X \text{ sub-Gaussian} \Rightarrow \left(\mathbb{E}[|x|^p] \right)^{1/p} \leq \overset{\text{there exists a universal } c_1 > 0}{c_1 \|X\|_{\psi_2} \sqrt{p}} \quad \forall p \geq 1$$

$$Y \text{ sub-Gaussian} \Rightarrow \left(\mathbb{E}[|y|^p] \right)^{1/p} \leq c_2 \|Y\|_{\psi_2} \sqrt{p} \quad \forall p \geq 1$$

$\rightarrow$ there exists a universal $c_2 > 0$

By the Cauchy-Schwarz inequality,

$$E[ZW] \leq \sqrt{E[Z^2] \cdot E[W^2]}$$

$$\begin{aligned} \left(E[|xy|^p] \right)^{1/p} &= \left(E[|x|^p |y|^p] \right)^{1/p} \\ &\leq \left(\left(E[|x|^{2p}] \right)^{1/2} \cdot \left(E[|y|^{2p}] \right)^{1/2} \right)^{1/p} \\ &= \left(E[|x|^{2p}] \right)^{1/2p} \cdot \left(E[|y|^{2p}] \right)^{1/2p} \\ &\leq c_1 \|x\|_{\psi_2} \cdot \sqrt{2p} \cdot c_2 \|y\|_{\psi_2} \sqrt{2p} \quad \forall p \geq 1 \\ &= 2c_1 c_2 \|x\|_{\psi_2} \|y\|_{\psi_2} p \quad \forall p \geq 1, \end{aligned}$$

from which the result follows.

⑦ If X is sub-Gaussian, then

$$\|X - E[X]\|_{\psi_2} \leq c_1 \|X\|_{\psi_2} \quad \text{for some universal } c_1 > 0,$$

and along similar lines, if X is sub-exponential, then

$$\|X - E[X]\|_{\psi_1} \leq c_2 \|X\|_{\psi_1} \quad \text{for some universal } c_2 > 0.$$

Proof of ⑦ :

For sub-Gaussian :

$$X \text{ sub-Gaussian} \Rightarrow E\left[e^{X^2 / \|X\|_{\psi_2}^2} \right] \leq 2$$

We then have

$$\begin{aligned} E\left[e^{\frac{(X-E[X])^2}{4\|X\|_{\psi_2}^2}} \right] &\leq E\left[e^{\frac{2(X^2 + (E[X])^2)}{4\|X\|_{\psi_2}^2}} \right] \\ &\quad \text{using } (a-b)^2 \leq 2a^2 + 2b^2 \\ &= E\left[e^{X^2 / (2\|X\|_{\psi_2}^2)} \right] \cdot e^{\frac{(E[X])^2}{2\|X\|_{\psi_2}^2}} \\ &\quad \text{Jensen's inequality } \leftarrow \begin{aligned} &E\left[\sqrt{e^{X^2 / \|X\|_{\psi_2}^2}} \right] \\ &\leq \sqrt{E\left[e^{X^2 / \|X\|_{\psi_2}^2} \right]} \end{aligned} \\ &\leq \sqrt{E\left[e^{X^2 / \|X\|_{\psi_2}^2} \right]} \cdot e^{\frac{(E[X])^2}{2\|X\|_{\psi_2}^2}} \end{aligned}$$

$$\leq \sqrt{2} \cdot e^{(E[X])^2 / (2 \|X\|_{\psi_2}^2)}$$

$$\stackrel{\substack{\text{T Jensen's} \\ \text{inequality} \\ h(x) = x^2}}{\leq} \sqrt{2} \cdot e^{E[X^2] / (2 \|X\|_{\psi_2}^2)}$$

$$\stackrel{\substack{\text{T Jensen's} \\ \text{inequality} \\ f(x) = e^x}}{\leq} \sqrt{2} \cdot E \left[e^{X^2 / 2 \|X\|_{\psi_2}^2} \right]$$

$$\stackrel{\substack{\text{T Jensen's} \\ \text{inequality} \\ g(x) = \sqrt{x}}}{\leq} \sqrt{2} \cdot \sqrt{E \left[e^{X^2 / \|X\|_{\psi_2}^2} \right]}$$

$$\leq \sqrt{2} \cdot \sqrt{2} = 2,$$

from which it follows that

$$\|X - E[X]\|_{\psi_2} \leq 2 \|X\|_{\psi_2}.$$

The proof for sub-exponential is along similar lines as above and is left as an exercise.

⑧ If X is sub-Gaussian, then

$$\|X - E[X]\|_{\psi_2}^2 \geq c \operatorname{Var}(X) \quad \text{for some universal } c > 0.$$

Proof of ⑧:

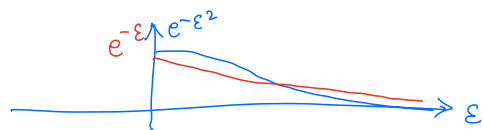
$$X \text{ sub-Gaussian} \Rightarrow X - E[X] \text{ sub-Gaussian}$$

$$\Rightarrow \left(E[|X - E[X]|^p] \right)^{1/p} \leq \overset{\substack{\text{there exists universal } c_2 > 0}}{c_2} \|X - E[X]\|_{\psi_2} \sqrt{p} \quad \forall p \geq 1$$

$$\stackrel{p=2}{\Rightarrow} \sqrt{\operatorname{Var}(X)} \leq c_2 \sqrt{2} \|X - E[X]\|_{\psi_2}$$

$$\Rightarrow \|X - E[X]\|_{\psi_2}^2 \geq \frac{1}{2c_2^2} \operatorname{Var}(X).$$

Bernstein's inequality



Fix $n \in \mathbb{N}$. Suppose that $X_1, \dots, X_n$ are independent and sub-exponential.

Fix $a_1, \dots, a_n \in \mathbb{R}$. Without loss of generality, let $E[X_k] = 0 \quad \forall k$.

Then,

$$\mathbb{P}\left(\left|\sum_{k=1}^n a_k X_k\right| \geq \varepsilon\right) \leq 2 \exp\left(-c \min\left\{\frac{\varepsilon^2}{\sum_{k=1}^n a_k^2 \|X_k\|_{\Psi_1}^2}, \frac{\varepsilon}{\max_k |a_k| \|X_k\|_{\Psi_1}}\right\}\right)$$

$c > 0$ is a universal constant

for small ε , decay is $e^{-O(\varepsilon^2)}$
for large ε , decay is $e^{-O(\varepsilon)}$

Corollary:

Let $X_1, \dots, X_n$ be independent and sub-Gaussian. Let $\underline{X} = (X_1, \dots, X_n)$.

Then, we know that

$$\|\underline{X}\|_2^2 = \sum_{k=1}^n X_k^2 \text{ is sub-exponential, and}$$

$$\mathbb{P}\left(\left|\|\underline{X}\|_2^2 - E[\|\underline{X}\|_2^2]\right| \geq \varepsilon\right) \leq 2 \exp\left(-c \min\left\{\frac{\varepsilon^2}{\sum_{k=1}^n \|X_k\|_{\Psi_2}^4}, \frac{\varepsilon}{\max_k \|X_k\|_{\Psi_2}^2}\right\}\right)$$

for some universal constant $c > 0$.

$\forall \varepsilon > 0$.

Application of the above Corollary to $\underline{X} \sim N(\underline{0}, \sigma^2 \mathbf{I})$.

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$. Let $\underline{X} = (X_1, \dots, X_n)$.

Then,

$$E[\|\underline{X}\|_2^2] = E\left[\sum_{k=1}^n X_k^2\right] = n\sigma^2.$$

Furthermore,

$$\|X_k\|_{\Psi_2} = \sqrt{\frac{8}{3}} \sigma^2 \quad \forall k \in \{1, \dots, n\}.$$

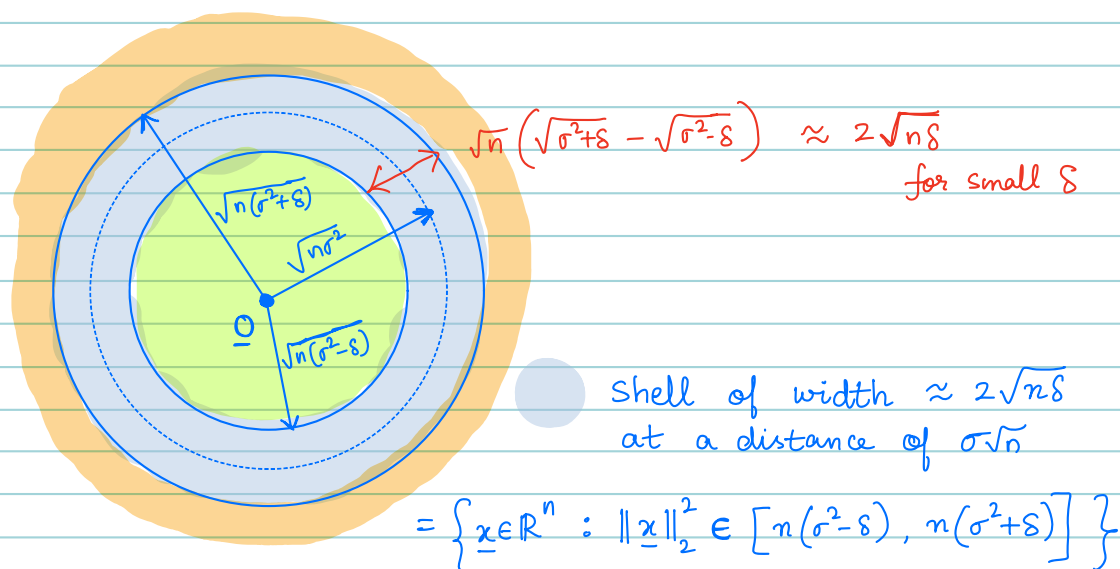
Applying the corollary, we get

$$\mathbb{P} \left(\left| \|x\|_2^2 - n\sigma^2 \right| \geq \overset{\varepsilon}{n\delta} \right) \leq 2 \exp \left(-c \min \left\{ \frac{n^2 \delta^2}{\sum_{k=1}^n \left(\frac{8}{3} \sigma^2 \right)^2}, \frac{n\delta}{\frac{8}{3} \sigma^2} \right\} \right)$$

$$\|x\|_2^2 \geq n(\sigma^2 + \delta)$$

$$\|x\|_2^2 \leq n(\sigma^2 - \delta)$$

$$= 2 \exp \left(-c' \min \left\{ \frac{n\delta^2}{\sigma^4}, \frac{n\delta}{\sigma^2} \right\} \right)$$



Inference: Most of the mass lies in the shell with high probability!

- It is intuitive to expect that

$$\mathbb{P} \left(\left| \|x\|_2^2 - n\sigma^2 \right| \geq n\delta \right) \leq 2 \exp \left(-O(n) \right)$$

- What is counterintuitive is that

$$\mathbb{P} \left(\left| \|x\|_2^2 - n\sigma^2 \right| \leq -n\delta \right) \leq 2 \exp \left(-O(n) \right).$$

The reason behind this is that

$$\text{Vol}_n(\text{green circle}) \ll \text{Vol}_n(\text{blue circle}) \quad \text{for large } n$$

this will be given as a guided exercise